CORINA KAISER

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NYU Master's candidate with background in linear algebra and metaphysics, now focused on AI and emergent agency through rapid self-directed learning, research, and experimentation.

EDUCATION

New York University, Tandon School of Engineering, Brooklyn, NY — May 2026
MS, Mathematical Sciences
- Inclusive Excellence in Mathematics Award

The Pennsylvania State University, University Park, PA — May 2022
BA, Mathematics | BA, Philosophy | BS, Applied German
- National German Honor Society (ΔΦΑ)
- Student Engagement Network Grant
    - Awarded to support research to improve upon the Growing Block Theory of time.

EXPERIENCE

LoveSmarter Lab, New York, NY
Research Assistant — Jan 2026—Present
- Manually review survey entries to distinguish human submissions from bots.
- Devise statistical models for bot detection by using Python to analyze metadata patterns and content coherency.
- Generate string variable recode syntax in IBM SPSS for data cleaning.

SUNY Brooklyn Educational Opportunity Center, Brooklyn, NY
Adjunct Lecturer — May 2023—Jun 2024
- Developed curriculum for instructing adult learners in various math and science courses.
- Courses taught: Math for Medical Assistants, Math for Food Service Workers, GED Math, GED Science

Math & English Tutor | Library Assistant — Dec 2022—May 2023
- Hosted tutoring sessions for adults working towards GEDs and Medical Assistant certifications
- Constructed and distributed thematic booklets to support students' learning of specific course concepts.

PROJECTS

Surrealist-Math Injection Experiment | brain.vat — Ongoing
Objective: Observe interactions between two autonomous GPT-2 models fine-tuned on complementary corpora.
- Developed online interface for continuous dialogue to investigate evolution of conversation topics.
- Currently retraining on alternative social logs to direct conversational style.
- Planned phase 2: nested CoT structure implemented on Qwen 0.5B.

NVIDIA Nemotron Model Reasoning Challenge — Ongoing
Objective: Fine-tune Nemotron-Nano-30B for improved reasoning on symbolic constraint puzzles.
- Synthesized 9,600 example CoT training set across 6 puzzle types.
- Trained 8-bit LoRA adapter (rank 32) via Kaggle on RTX Pro 6000 Blackwell; competition score: 0.66 (above 2,136 submission leaderboard median)
- Patched broken Blackwell toolchain to allow training under offline competition constraints.

WearabLLM — April 2026
Objective: Provide LLM an embodied interface to allow multimodal expression.
- Developed iOS app handling audio input, LLM API calls, and Bluetooth communication with Adafruit board.
- Implemented serial command protocol to allow LLM to drive LED, audio, and servo motor output.

SKILLS

- **Languages:** JavaScript, Python (working fluency), SQL, SPSS, German
- **Hardware:** Arduino/embedded C, CircuitPython, BLE communication, e-textiles, 3D printing
- **Data Infrastructure:** HDFS, Spark, Parquet, MapReduce (MRJob), Dataproc, query optimization
- **ML/DL:** Clustering/regression, PCA, PyTorch, HuggingFace Transformers, PEFT, LoRA/QLoRA